

pH of 0.1M $\text{Na}_2\text{S}(\text{aq})$ is :

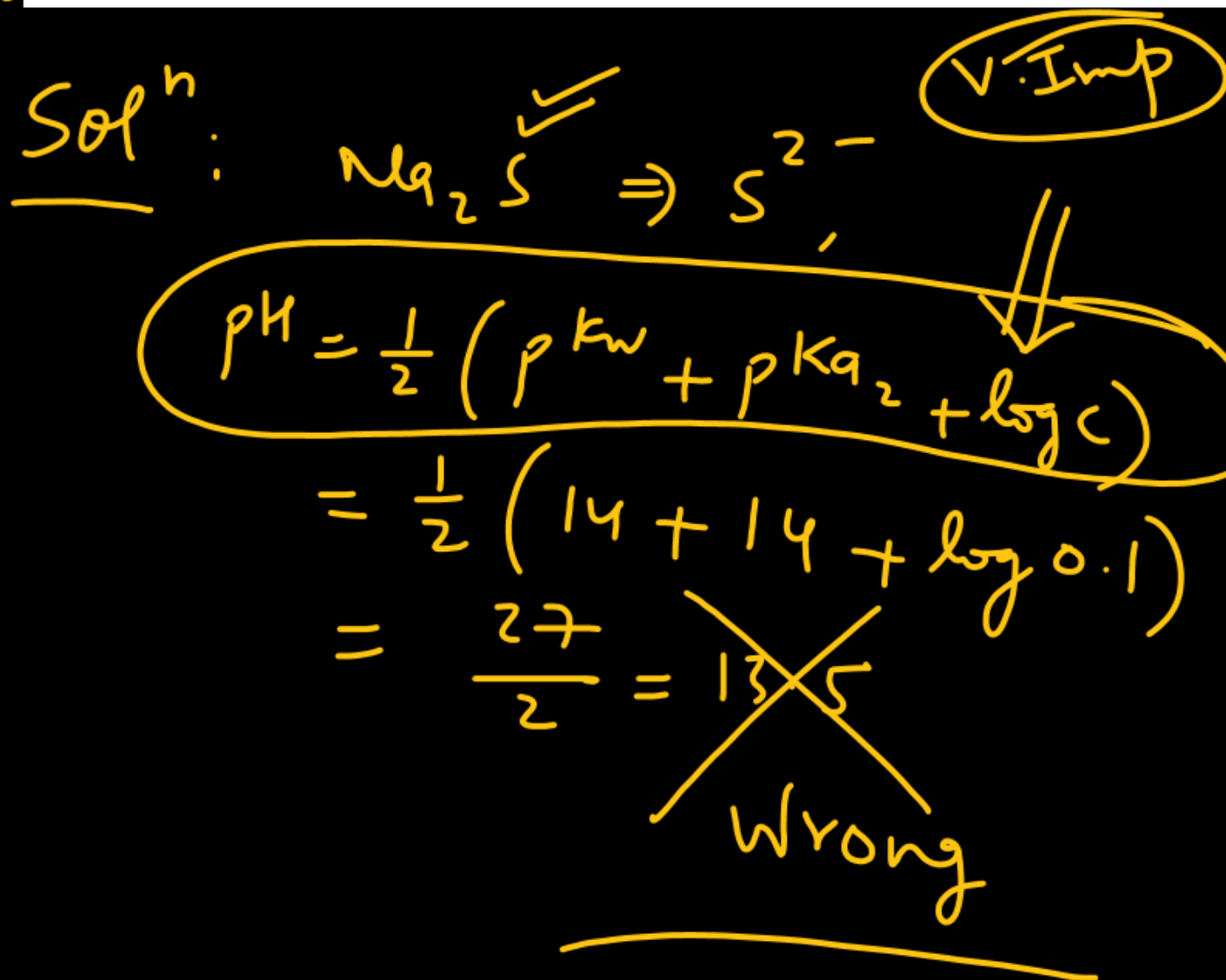
$[\text{K}_{\text{a}1} = 10^{-7}, \text{K}_{\text{a}2} = 10^{-14} \text{ for } \text{H}_2\text{S}]$

✓ (A) 13

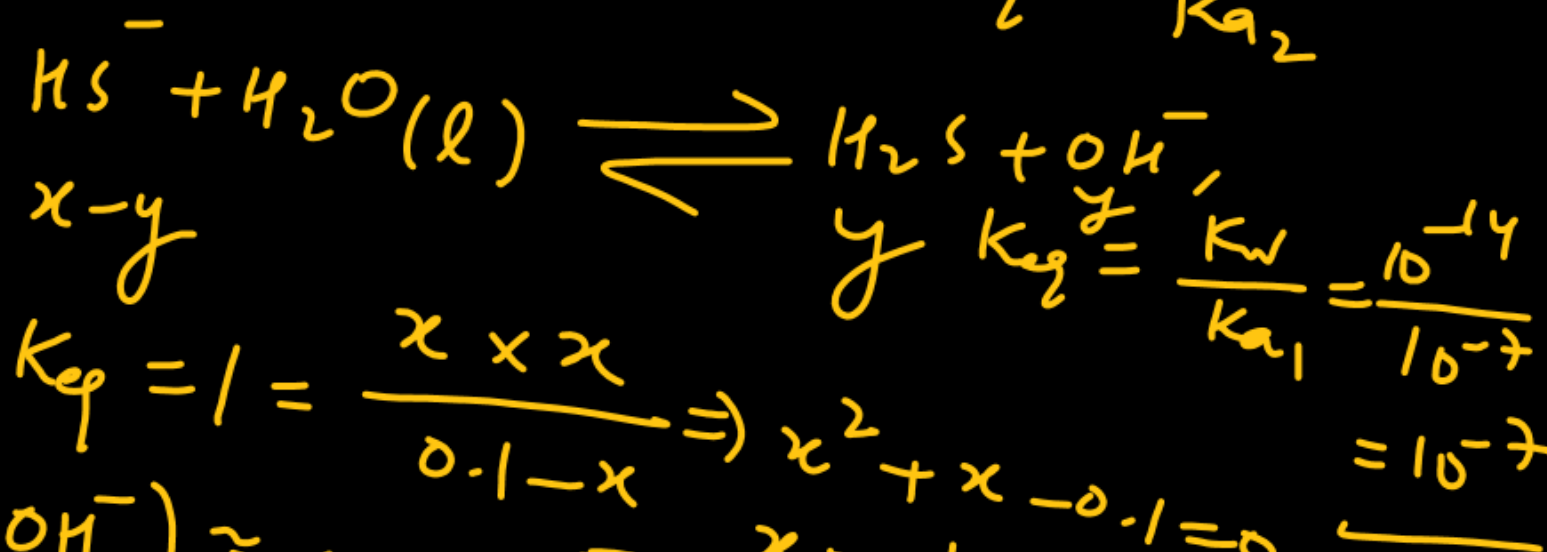
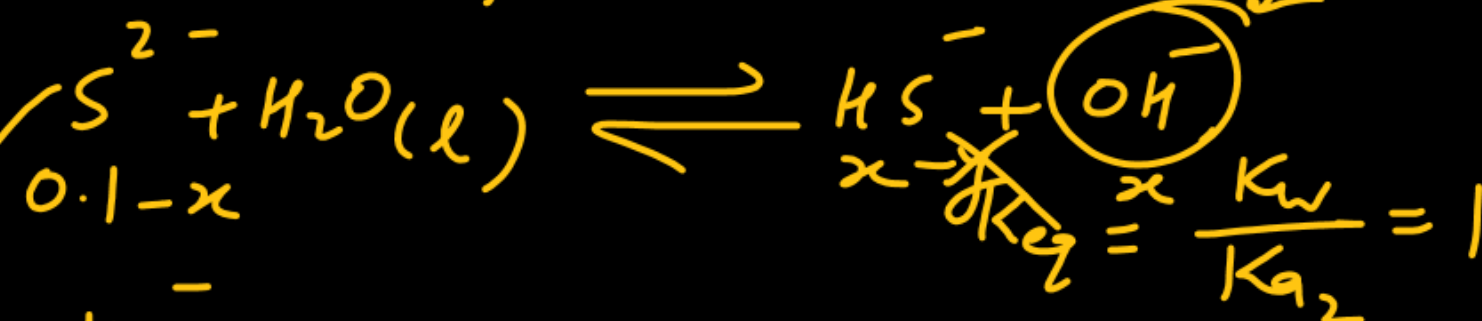
(B) 13.5

(C) 1

(D) 0.5



Correct approach:



$\text{K}_{\text{a}2} = 1 = \frac{x \times x}{0.1-x} \Rightarrow x^2 + x - 0.1 = 0$

$x = \frac{-1 + \sqrt{1+0.4}}{2}$

$\approx \frac{1.2-1}{2} = 0.1\text{M}$

$[\text{OH}^-] \approx 0.1\text{M}$

$\text{pOH} = 1$

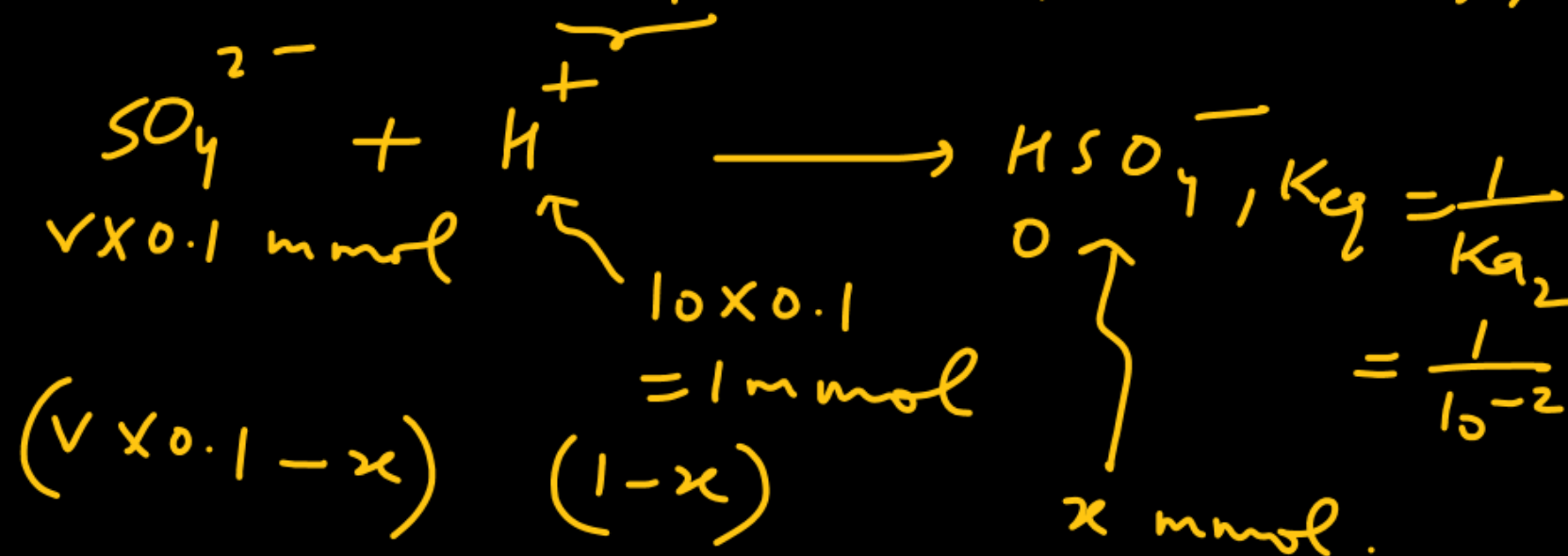
$\text{pH} = 14 - 1 = 13$ Ans

Calculate the volume of 0.1M $\text{Na}_2\text{SO}_4(\text{aq})$ which must be added to 10 ml of 0.1M $\text{HCl}(\text{aq})$ so that pH of resulting solution becomes 2.

[$K_{a1}(\text{H}_2\text{SO}_4)$ = very large, $K_{a2}(\text{H}_2\text{SO}_4) = 10^{-2}$]

Ans: 15 ml

Solⁿ: $V \text{ ml}, 0.1\text{M } \text{Na}_2\text{SO}_4 + 10 \text{ ml}, 0.1\text{M } \text{HCl}(\text{aq})$.



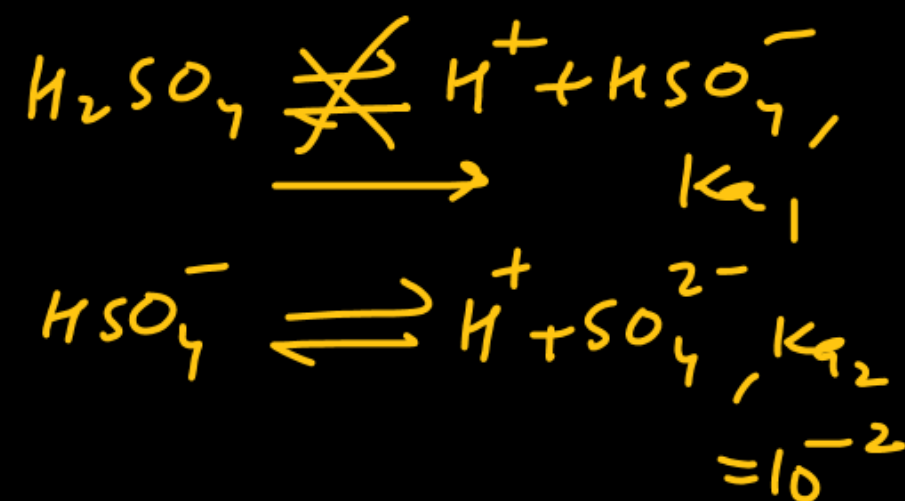
$$(\text{H}^+)_f = 10^{-2} \text{ M}$$

$$\frac{1-x}{V+10} = 10^{-2} \text{ M}$$

$$\frac{1-0.05V}{V+10} = \frac{1}{100}$$

$$100 - 5V = V + 10$$

$$6V = 90 \Rightarrow V = 15 \text{ ml}$$



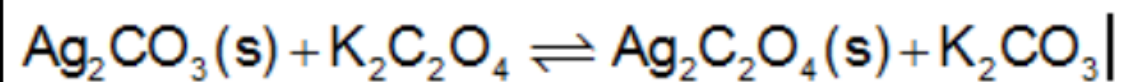
$$K_{a2} = 10^{-2} = \frac{(x/(V+10)) \times 10^{-2}}{(0.1V-x)/(V+10)}$$

$$1 = \frac{x}{0.1V-x}$$

$$0.1V - x = x$$

$$x = 0.05V$$

The solubility product of $\text{Ag}_2\text{C}_2\text{O}_4$ at 25°C is 1.3×10^{-12} . A solution of $\text{K}_2\text{C}_2\text{O}_4$ containing 0.15 moles in 500 ml water is shaken at 25°C with excess Ag_2CO_3 till the following equilibrium is reached.



At equilibrium, the solution contains 0.02 mole of K_2CO_3 . Calculate K_{sp} of Ag_2CO_3 .

Ans : $K_{\text{sp}}(\text{Ag}_2\text{CO}_3) = 2 \times 10^{-13}$

$K_{\text{sp}} = 2 \times 10^{-13}$

Solⁿ: $K_{\text{sp}}(\text{Ag}_2\text{C}_2\text{O}_4) = 1.3 \times 10^{-12}$

(Precipitation exchange)

500ml
0.5L



0.15 mol

0.15 - x

→ 0.02 mol.

= 0.13 mol

x mol

x mol.

= 0.02 mol

$$K_{\text{eq}} = \frac{[\text{CO}_3^{2-}]}{[\text{C}_2\text{O}_4^{2-}]} = \frac{[\text{Ag}^+]^2 \times [\text{CO}_3^{2-}]}{[\text{Ag}^+]^2 \times [\text{C}_2\text{O}_4^{2-}]} \Rightarrow$$

$$\frac{0.02/0.5}{0.13/0.5} = \frac{K_{\text{sp}}(\text{Ag}_2\text{CO}_3)}{K_{\text{sp}}(\text{Ag}_2\text{C}_2\text{O}_4)}$$

$$\frac{2}{13 \times 10} = \frac{K_{\text{sp}}(\text{Ag}_2\text{CO}_3)}{1.3 \times 10^{-12}}$$

10 mL of 0.1 M tribasic acid H_3A is titrated with 0.1 M NaOH solution. What is the ratio of $\frac{[H_3A]}{[A^{3-}]}$ at 2nd equivalence point? [Given : $K_{a1} = 10^{-3}$, $K_{a2} = 10^{-8}$, $K_{a3} = 10^{-12}$]

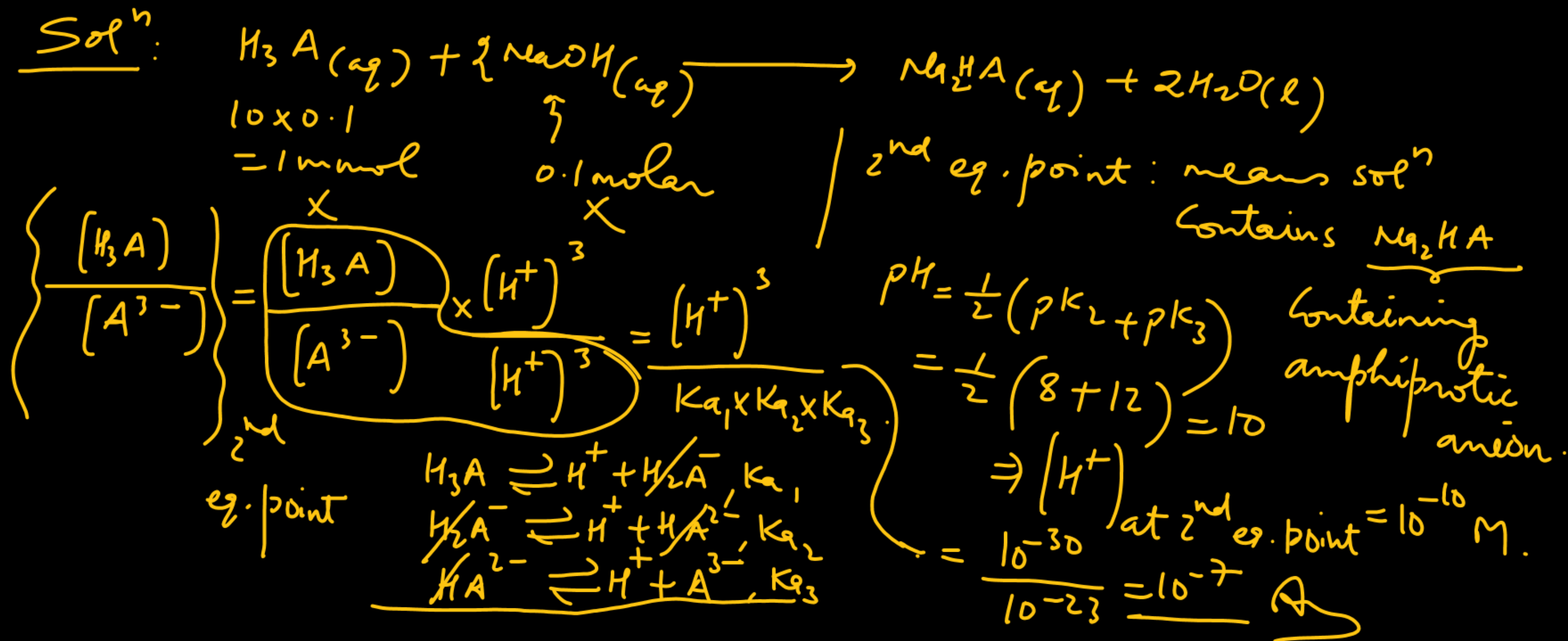
(A) 10^{-2}

(B) 10^{+4}

✓ (C) 10^{-7}

(D) 10^{-6}

good



Which of the following mixtures can act as a buffer ?

(A) NaOH + HCOONa (1 : 1 molar ratio) ~~(B) HCOOH + NaOH (2 : 1 molar ratio)~~

~~(C) NH₄Cl + NaOH (2 : 1 molar ratio)~~ (D) HCOOH + NaOH (1 : 1 molar ratio)

TEE-Adv

Solⁿ:

(A) ^{base} NaOH + ^{basic solⁿ} HCOONa (Not a buffer)

HCOOH: acid

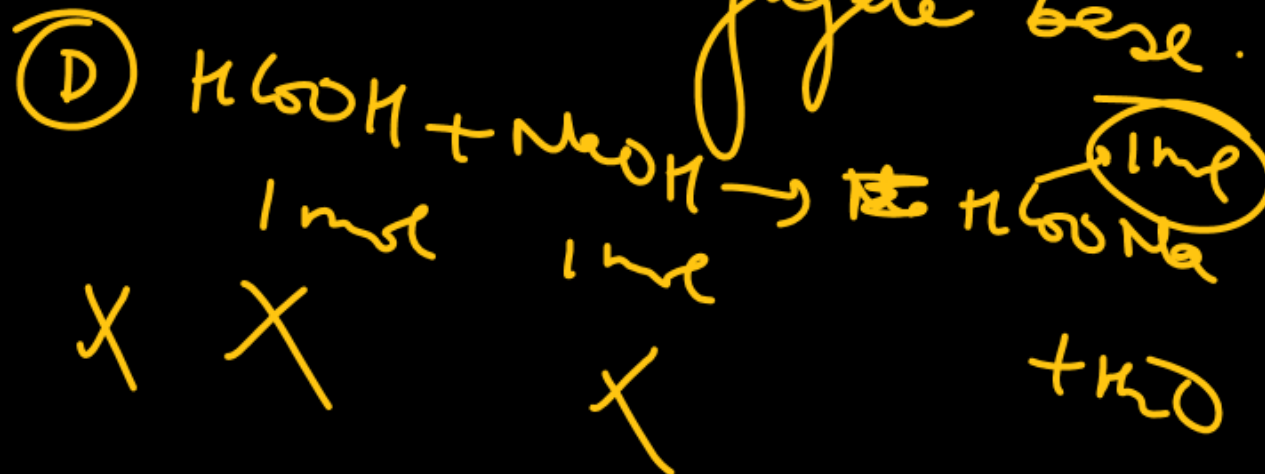
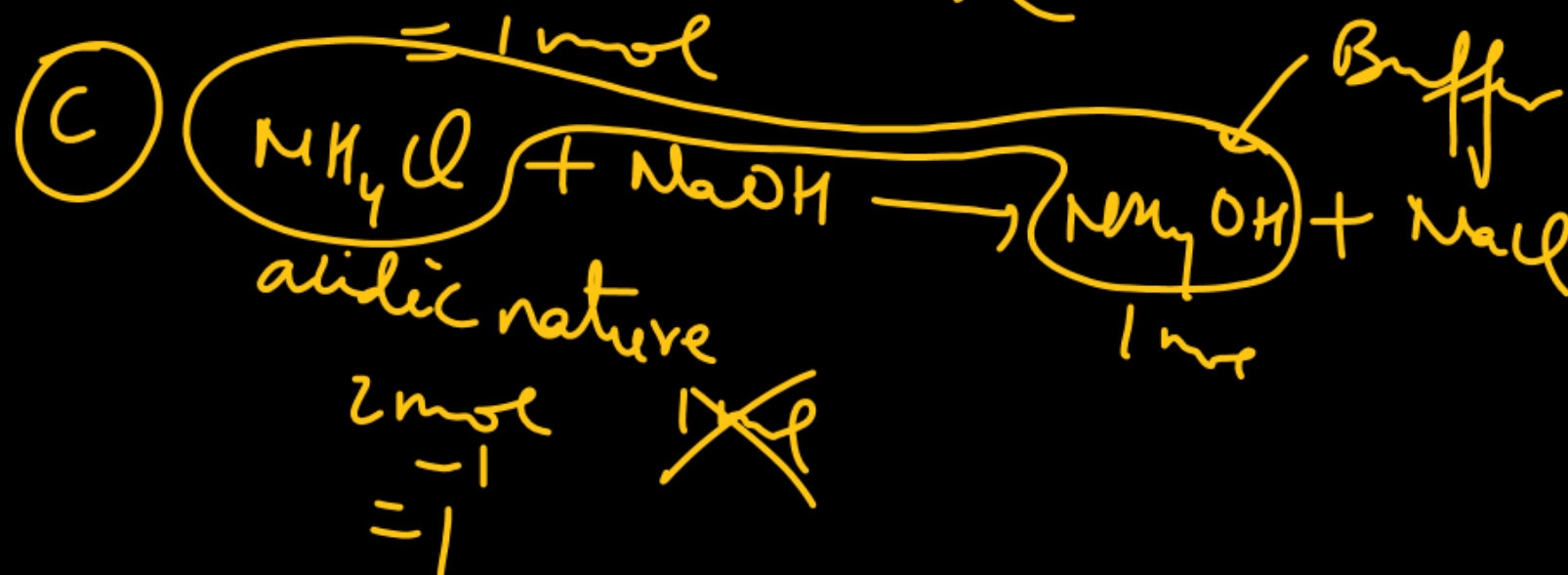
HCOO⁻: base (conjugate)



Buffer.

HCOOH + HCOONa

WA Conjugate base.



Which is/are correct statement (s) about the solubility of AgCl(s) ?

Given : $K_{sp}(\text{AgCl}) = 10^{-10}$; $K_f[\text{Ag}(\text{NH}_3)_2^+] = 10^8$

(A) Solubility of AgCl in pure water is 10^{-5} gm/litre X

(B) Solubility of AgCl in 2M KBr is 10^{-5} mol/litre X or ✓

(C) Solubility of AgCl in 2M AgNO_3 is 5×10^{-11} M ✓

(D) Solubility of AgCl in 2M NH_3 is 0.182 M X

Ans : B, C

Solⁿ: (A) $\text{AgCl(s)} \Rightarrow s = \sqrt{K_{sp}} = 10^{-5} \text{ mol/lit} \Rightarrow$

(B) $\text{AgCl(s)} + \text{Br}^- \rightarrow \text{AgBr} \downarrow + \text{Cl}^-$
yellow

(C) $\text{AgCl(s)} \rightleftharpoons \text{Ag}^+ + \text{Cl}^-$
 $(s+2)s$
 (common ion effect)

$K_{sp} = \frac{[\text{Cl}^-]}{[\text{Br}^-]} \times \frac{[\text{Ag}^+]}{[\text{Ag}^+]} = \frac{K_{sp}(\text{AgCl})}{K_{sp}(\text{AgBr})}$
 $(s+2) \times s = 10^{-10}$
 $s = 5 \times 10^{-11} \text{ M}$

(D) $\text{AgCl(s)} \rightleftharpoons \text{Ag}^+ + \text{Cl}^-$, K_{sp}
 $\text{Ag}^+ + 2\text{NH}_3 \rightarrow [\text{Ag}(\text{NH}_3)_2]^+$, 10^8

$K_f = \frac{s}{[\text{Ag}^+](2-2s)^2}$
 $10^8 = \frac{s}{\frac{10^{-10}}{s} \times (2-2s)^2}$
 $10^{-2} = \left(\frac{s}{2-2s}\right)^2 \Rightarrow \frac{s}{2-2s} = 0.1$
 $s = \frac{1}{6} = 0.167 \text{ M}$

$K_{sp}(\text{AgBr}) < K_{sp}(\text{AgCl})$
 $10^{-14} < 10^{-10}$

$\frac{K_{sp}(\text{AgCl})}{K_{sp}(\text{AgBr})} = \frac{10^{-10}}{10^{-14}} = 10^4$
 actual data

In an acidic indicator HIn has ionisation constant is 10^{-8} . The acid form of indicator is yellow and alkaline form is red. Which is correct statement :

(Given : $\log 2 = 0.3$, $\log 3 = 0.48$)

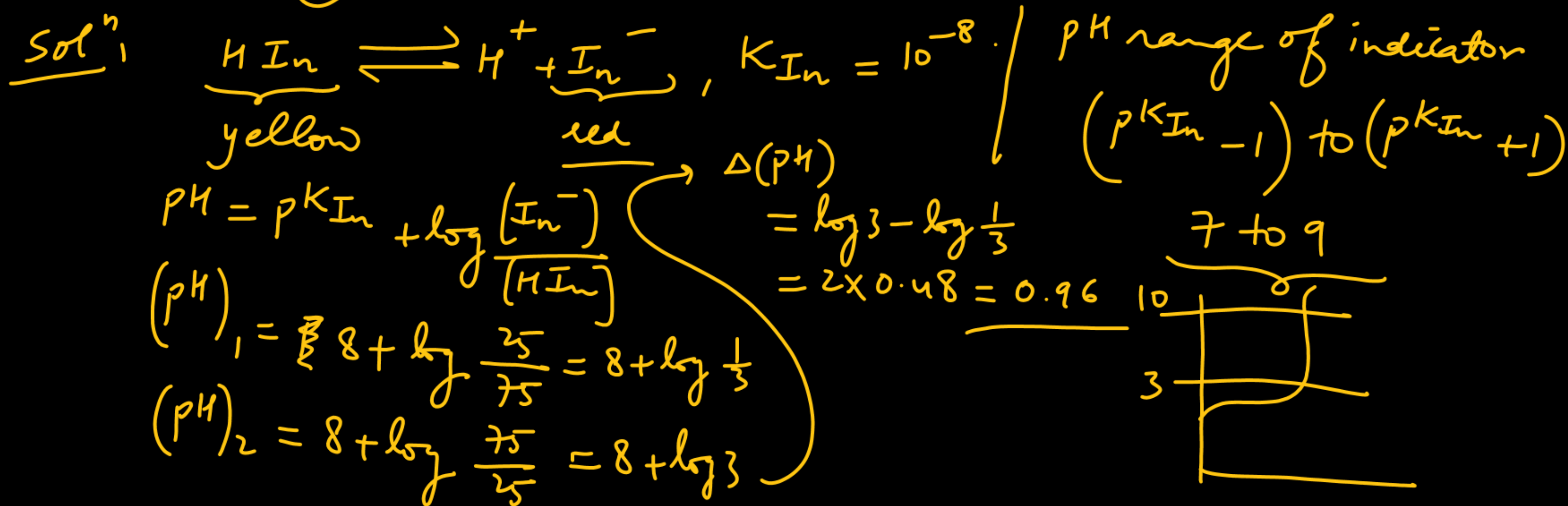
- ✓ (A) The pH range of indicator is 7 to 9 ✓
- ✓ (B) Change in pH is 0.96 when 75% yellow colour change to 75% red colour ✓
- ✓ (C) This indicator is suitable for the titration of strong acid Vs strong base ✓
- (D) pH of indicator is 8.3 when ratio of acid form to alkaline form is 2. ✗

$$\textcircled{D} \quad \frac{[HIn]}{[In^-]} = \frac{2}{1}$$

$$pH = 8 + \log \frac{1}{2}$$

$$= 8 - 0.3$$

$$= 7.7$$



Column-I	Column-II
(A) CH_3COOH ($\text{pK}_a = 4.74, 0.1\text{M}$) + CH_3COONa (0.1M)	(p) Acidic buffer at its maximum capacity
(B) CH_3COOH (0.1M) + HCl (0.1M)	(q) Buffer solution
(C) CH_3COOH ($\text{pK}_a = 4.74, 0.1\text{M}$) + NH_4OH ($\text{pK}_b = 4.74, 0.1\text{M}$)	(r) $\text{pH} < 7$ at 25°C
(D) CH_3COONa (300 mL of 0.1M) + HCl (100 mL of 0.1M)	(s) $\text{pH} = 7$ at 25°C

Solⁿ: (A) CH_3COOH (0.1M) + CH_3COONa (0.1M) (Conjugate acid base pair)
Buffer, $\text{pH} = \text{pK}_a + \log \frac{s}{a} = 4.74 < 7$

(B) → 2, 1, 2

(C) CH_3COOH (0.1M) + NH_4OH (0.1M) → $\text{CH}_3\text{COONH}_4$ ($0.1 \times V \text{ mol}$) + H_2O .
Buffer solⁿ.

(D) CH_3COONa ($300 \times 0.1 = 30 \text{ mmol}$) + HCl ($10 \times 0.1 = 10 \text{ mmol}$) → CH_3COOH (10 mmol) + NaCl (10 mmol)
 $= 20$

$$\text{pH} = 4.74 + \log \frac{20}{10} = 4.74 + 0.3 < 7$$

(D) → 1, 2, 1

$$\text{pH} = \frac{1}{2} (\text{pK}_a + \text{pK}_b) = \frac{1}{2} (4.74 + 4.74) = 7$$

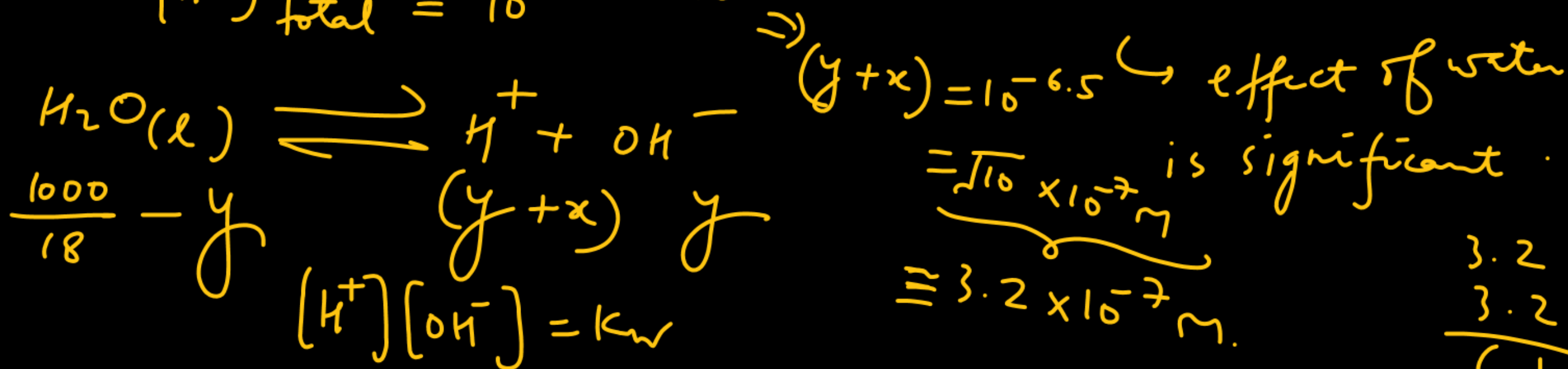
pH of an aqueous solution of x molar HClO_4 at 25°C is 6.5. Find x

Solⁿ:

HClO_4 : strong acid

$$[\text{H}^+]_{\text{total}} = 10^{-\text{pH}} = 10^{-6.5} \text{ M}$$

$$6 < (\text{pH})_{\text{seen}} < 8$$



$$\sqrt{10} \times 10^{-7} \times y = 10^{-14} \Rightarrow y = \frac{1}{\sqrt{10}} \times 10^{-7} \text{ M}$$

$$x = [\text{HClO}_4] = \sqrt{10} \times 10^{-7} - \frac{1}{\sqrt{10}} \times 10^{-7} = \frac{9}{\sqrt{10}} \times 10^{-7} \text{ M} \approx 2.8 \times 10^{-7} \text{ M}$$

$$\begin{array}{r} 3.2 \\ 3.2 \\ \hline 6.4 \\ 9.6 \\ \hline 10.24 \end{array}$$

Amongst the following, the total number of compounds whose aqueous solution turns red litmus paper blue is :

NaCN	KCl	CH ₃ COONH ₄	NaH ₂ PO ₄	ZnCl ₂	Na ₃ PO ₄
Fe(NO ₃) ₃	Na ₂ CO ₃	NH ₄ Cl	NaHCO ₃	Na ₂ C ₂ O ₄	Na ₂ HPO ₄

Given :

Acid	Ka ₁	Ka ₂	Ka ₃
H ₃ PO ₄	10 ⁻³	10 ⁻⁸	10 ⁻¹²
H ₂ CO ₃	10 ⁻⁶	10 ⁻¹¹	—
H ₂ C ₂ O ₄	10 ⁻²	10 ⁻⁵	—